

Assignment 2

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1

1.a

The starting address of the array is 420.

Since each element occupies 4 storage units, then going back 20 steps of 4 (80) gets to 420

1.b

Stacks and queues both hold data and have the same push/pop/peek functions, the only difference is the order in which data is popped.

1.c

The root is at level 0, which means including the node there will be 9 levels.

Assuming that each node (including the root) only has one child, then there would be only 9 nodes in the tree.

Assuming that each node has both children, then each level would add a power of two to the total count:

root: 1

1st level: 2

2nd level: 4

...

8th level: 256

total -> $1 + 2 + 4 + \dots + 256 = 2^9 - 1$

1.d

The tree must have a depth of at least 7, because a complete binary tree of depth 8 can only have 255 nodes. It could be a complete binary tree up to depth 6 and only have one node on the 8th level, or trade some more nodes from other levels and place them at the bottom (either level 7, or it could just keep going deeper). Either way it would have 128 leaf nodes (full 6th level is 128 nodes, but one node has one child which doesn't change the leaf count, and every other node in the 7th layer would take one away from elsewhere, so the leaf count will always be 128)

1.e

Stack input: 1, 2, 3, 4, 5, 6

Possible outputs:

- 1, 2, 3, 4, 5, 6 (push each element before adding the next)
- 3, 2, 1, 6, 5, 4 (push three, pop three, push the rest, pop the rest)
- 2, 1, 4, 3, 6, 5 (push two, pop two, repeat)

2

2.a In-order

order: 5, 10, 12, 17, 9, 3, 7, 25, 23, 11, 27, 33, 19, 6, 8

2.b Pre-order

order: 25, 9, 12, 10, 5, 17, 7, 3, 33, 10, 23, 27, 19, 8, 6

2.c Post-order

order: 5, 10, 17, 12, 3, 7, 9, 23, 27, 11, 6, 8, 19, 33, 25

3

3.a

postfix: $75/G6^*C*A3^*B^*+Z+T-$

3.b

postfix: $XY2^*7A2^*B2^*/1-12/^/*Z-$

3.c

prefix: $\frac{X}{Y-Z} - \frac{A}{8(0-\frac{P}{Q}) \cdot G^2}$

3.d

prefix: $b - 2 + \frac{4-ac}{2a} + 7c$